variety of tools available to help run a PKI for you. For example, Cloudflare's PKI toolkit (https://cfssl.org) and Hashicorp Vault (https://www.vaultproject.io/docs/secrets/pki/index.html) can both be used to automate most aspects of running a PKI. These general-purpose tools still require a significant amount of effort to integrate into a Kubernetes environment. An alternative that is becoming more popular in recent years is to use a *service mesh* such as Istio (https://istio.io) or Linkerd (https://linkerd.io) to handle TLS between services in your cluster for you.

> DEFINITION A ***service mesh*** is a set of components that secure communications between pods in a cluster using proxy sidecar containers. In addition to security benefits, a service mesh provides other useful functions such as load balancing, monitoring, logging, and automatic request retries.

A service mesh works by installing lightweight proxies as sidecar containers into every pod in your network, as shown in figure 10.6. These proxies intercept all network requests coming into the pod (acting as a reverse proxy) and all requests going out of the pod. Because all communications flow through the proxies they can transparently initiate and terminate TLS, ensuring that communications across the network are secure while the individual microservices use normal unencrypted messages. For example, a client can make a normal HTTP request to a REST API and the client's service-mesh proxy (running inside the same pod on the same machine) will transparently upgrade this to HTTPS. The proxy at the receiver will handle the TLS connection and